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Universal Airlines

Topic	Difficulty	Style
Market entry Market sizing	Advanced	Interviewer-led (McKinsey style)

Problem definition

Our client is Universal Airlines that offers regular passenger flight in economy, business & first class. They have been doing quite well and profits were strong so they build up **cash reserves** that they are now looking to invest. The company is thinking about investing in the **MRO business (Maintenance, Repair & Overhaul)** for global **business jets**.

The client has asked us to help him size up the market.

Comments

This case is made to be **interviewer-led**. Therefore the interviewer should guide the interviewee through the interview.

The case is split into **two parts**.

The **first part** describes more **qualitative problems** and has more **open questions** that should make the interviewee think about the problem and its solution. The **second part** is more about **quantitative problems** and calculations. Here the interviewee should try to make his own **calculations** and solve the questions.

Short Solution

The market size is around **\$620 m** for roughly **4,000 business jets** who receive 3 light checks each year and one heavy check every 3 years.

Paragraphs highlighted in green indicate diagrams or tables that can be shared in the “Case exhibits” section.

Paragraphs highlighted in blue can be verbally communicated to the interviewee.

The following framework/structure provides an overview of the case:

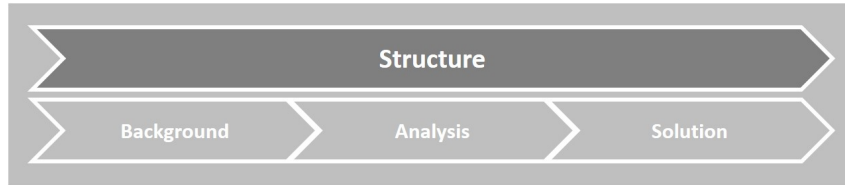


Diagram 1 - Case structure

I. Background

1. How would you go about structuring the analysis?

The case requires a **three-step process** including the understanding of the MRO business, segmentation of the market & estimating market size. Candidates are free to study more aspects or have broader structure, but interviewer should try to keep the structure **centered around these 3 steps**.

2. How do you think the MRO business works and what kind of services/products do you think are offered?

Once the candidate gives his response, interviewer can share **additional information** with the candidate so that he has a clear view of the business.

Share Table 1 with the candidate.

Additional information that can be shared:

- Maintenance of aircrafts is either done during **pre-flight, during the light check or the heavy check**.
- A **second** service component is the overhaul which includes **retro-fit of the aircraft layout or cleaning & repairing** carpets and plane seats.

- These cleansing & overhaul services are **typically done during light & heavy checks**.

3. Why does the airline choose to have the cleansing & overhaul services performed during light & heavy checks?

Possible answer:

To **save time & money**, maintenance providers typically carry out multiple MRO tasks in parallel so that they don't have to ground the aircraft when there is no need to.

II. Analysis

4. Now that you understand the business, what would be the next step?

The candidate should now look into segmenting the market and candidate should ask for information on this and the proper way to segment the market. Interviewer should challenge candidate to come up with his way of segmentation before sharing info with candidate.

Share Table 2 with the candidate.

Candidate should note that the business-jet segment is **quite dominant** in terms of numbers and that this validates the client's focus on that particular segment.

5. Can you think of a reason why it's also possible to focus on the wide-bodys instead of the business jets and why that's not very beneficial in the case of our client?

Possible answer:

The widebodys could generate **more revenue**, since their size is much bigger than the business jets. However, given their limited numbers it would be **difficult to enter** this segment, since the market might be saturated with existing players.

A **second** reason is that servicing widebodies would mean a much **larger capital investment** in terms of materials & facilities due to their size.

6. Makes sense, let's move on to sizing the market. How would you approach this?

Candidate should indicate that in order to size the market, we would need to estimate total revenues for the business jet segment. Important revenue drivers are **labor costs, hours of labor required & cost of materials** & potentially some overhead.

Candidate should also ask for data and information on those aspects.

Share Table 3 with the candidate.

Additional information that can be shared:

- Main revenue drivers are labor cost, manhours and cost of materials.
- Overhead can be left out for now, since those costs are minimal.
- Also, line maintenance should be outside the scope since most airlines have their own onboard engineers to conduct pre-flight checks.
- That leaves with the light & heavy checks only to take into account.

7. What would our estimated market size look like?

Labor costs:

- **Light check:** 180 hours*\$70=\$12,600.
- **Heavy check:** 1.300 hours*\$130=\$169,000.

Material costs:

- **Light check:** 10% of total revenues (labor+materials) makes up for the cost of materials which is \$1,400: $\text{Material costs} = 10\% * (12,600 + \text{material costs})$
- **Heavy check:** 50% of total revenues (labor+materials) makes up for

the cost of materials which is \$169,000.

Total revenue:

- **Light check:** $\$12,600 + \$1,400 = \$14,000$.
- **Heavy check:** $\$169,000 + \$169,000 = \$338,000$.

Total market size:

- Per year there are **3 light checks** and **1/3 of a heavy check** so we can calculate total size as:
 - $(3 \times \$14,000) + (1/3 \times \$338,000) = \$155,000$.
 - Given that we have roughly 4,000 business jets, our total annual market size is: **$155,000 \times 4,000 = \$620 \text{ m}$** .
- It's possible to multiply this number by the average life-cycle of an average business-jet, but no data is available on that.

III. Solution

The market size is around **\$620 m** for roughly **4,000 business jets** who receive 3 light checks each year and one heavy check every 3 years.

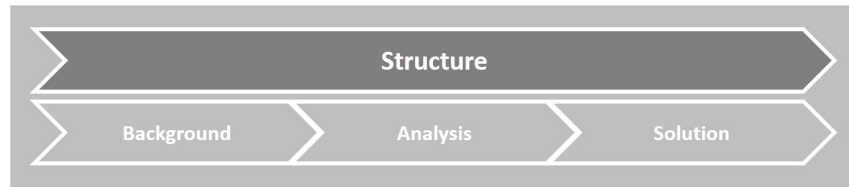


Diagram 1 - Case structure

1. How would you go about structuring the analysis?

2. How do you think the MRO business works and what kind of services/products do you think are offered?

MRO Segments	Description
Maintenance	▪ Light maintenance: Involves pre-flight checks ▪ Light checks: Inspection of major components every 4 months. ▪ Heavy-checks: In-depth inspection of entire aircraft every 3 years.
Repair	▪ Repair of damages during maintenance periods.
Overhaul	▪ Entire retro-fit of of interior including carpets, plane seats etc.

Table 1 – MRO Business

3. Why does the airline choose to have the cleansing & overhaul services performed during light & heavy checks?

4. Now that you understand the business, what would be the next step?

# of aircraft in service	A	B	C	D	E	F	Total
Business liner - widebody	9	47	0	0	0	0	56
Business liner- narrowbody	23	171	0	0	0	0	194
Business jet	0	0	1,555	1,128	96	1,179	3,958
Total	32	218	1,555	1,128	96	1,179	4,208

Table 2 – Aviation sector segmentation

5. Can you think of a reason why it's also possible to focus on the wide-bodys instead of the business jets and why that's not very beneficial in the case of our client?

6. Makes sense, let's move on to sizing the market. How would you approach this?

Type of service	Hours required	Hourly cost (\$)	Material cost
Light check (4 months)	180 hours	70	10% of total revenue
Heavy check (3 years)	1.300 hours	130	50% of total revenue

Table 3 – Revenue drivers

7. What would our estimated market size look like?